Testing Reddening Estimates with SNe Ia

ACME
Astrophysics Centre for Multimessenger studies in Europe



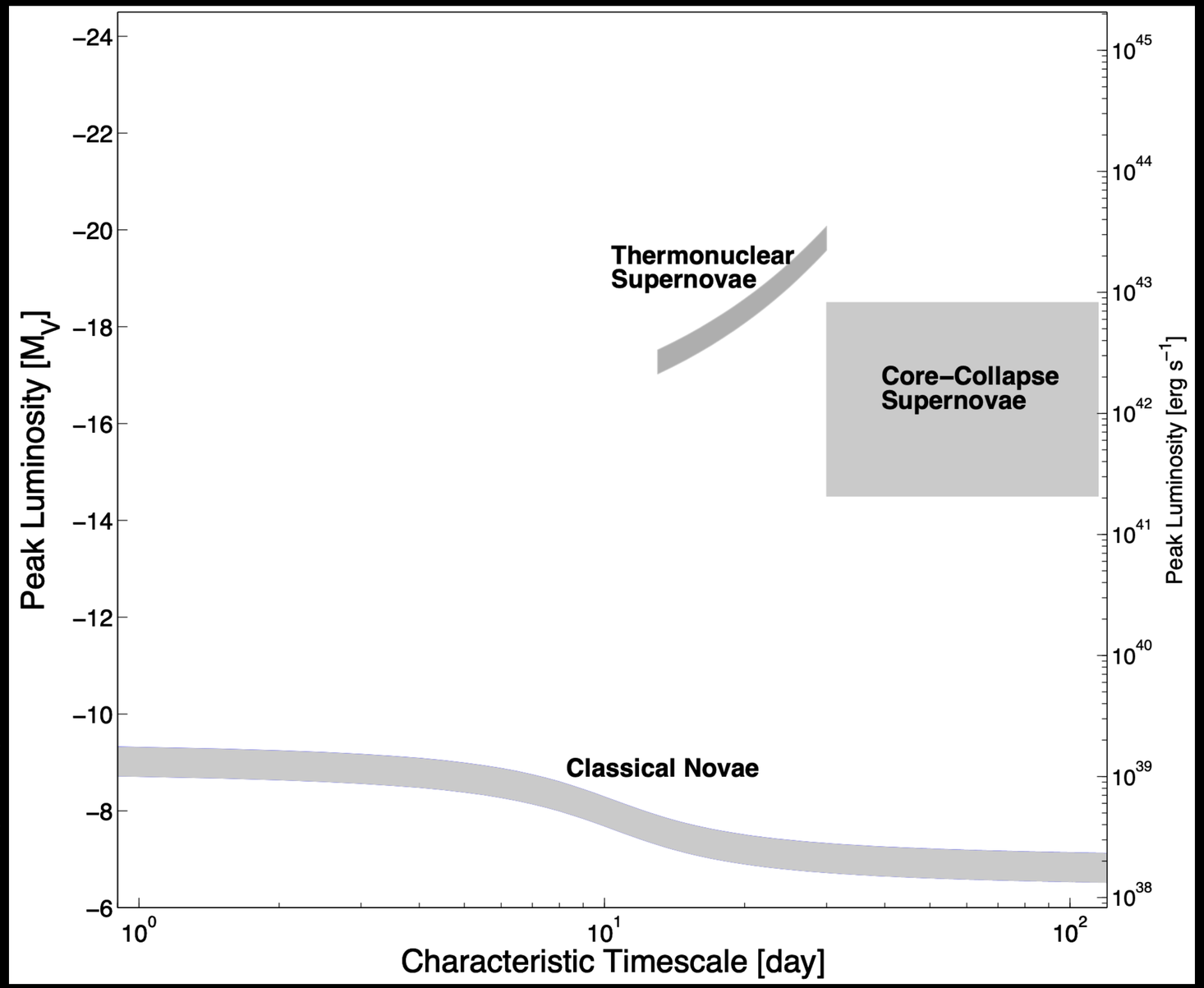
Giorgio Valerin, Asiago, 2026-10-12

Thermonuclear SNe (Type Ia)

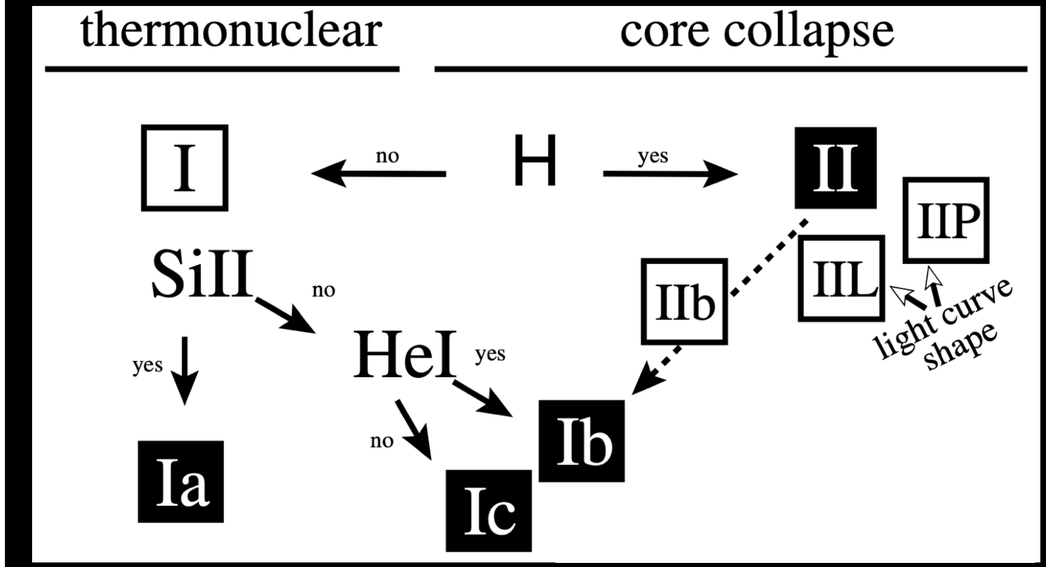




The variety within the transient sky

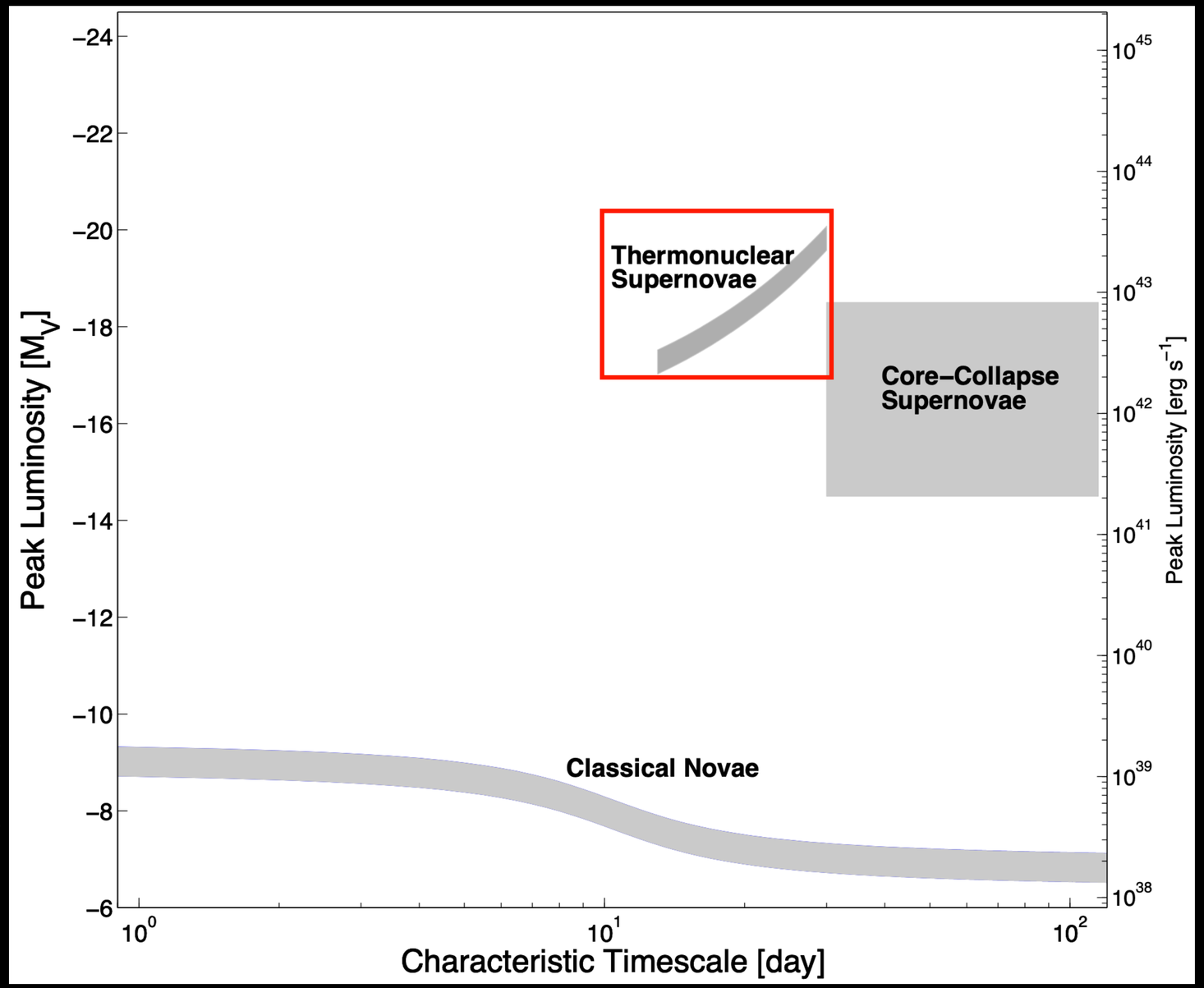


Kasliwal (2011)

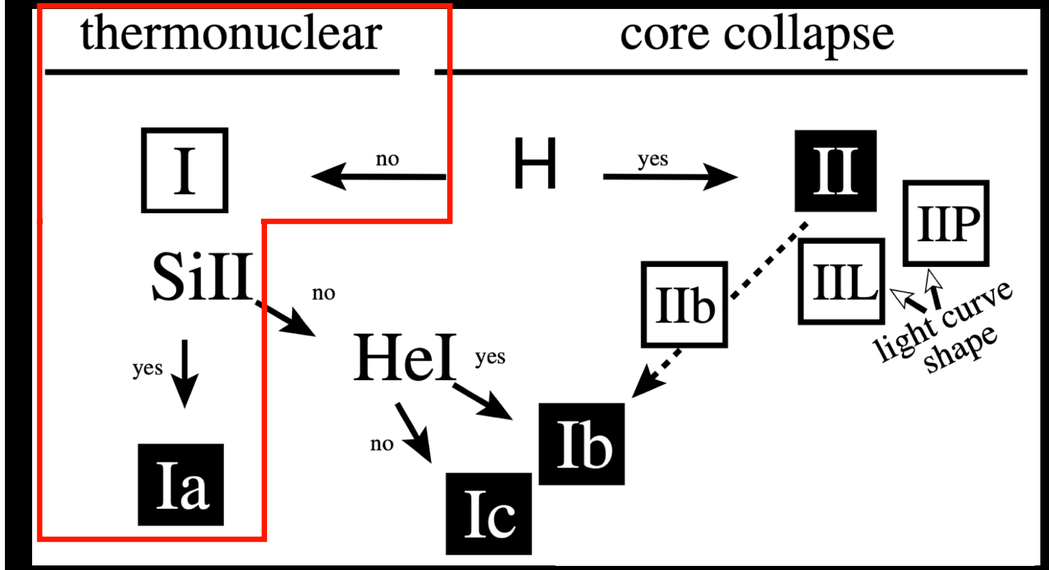


Adapted from Turatto (2003)

The variety within the transient sky

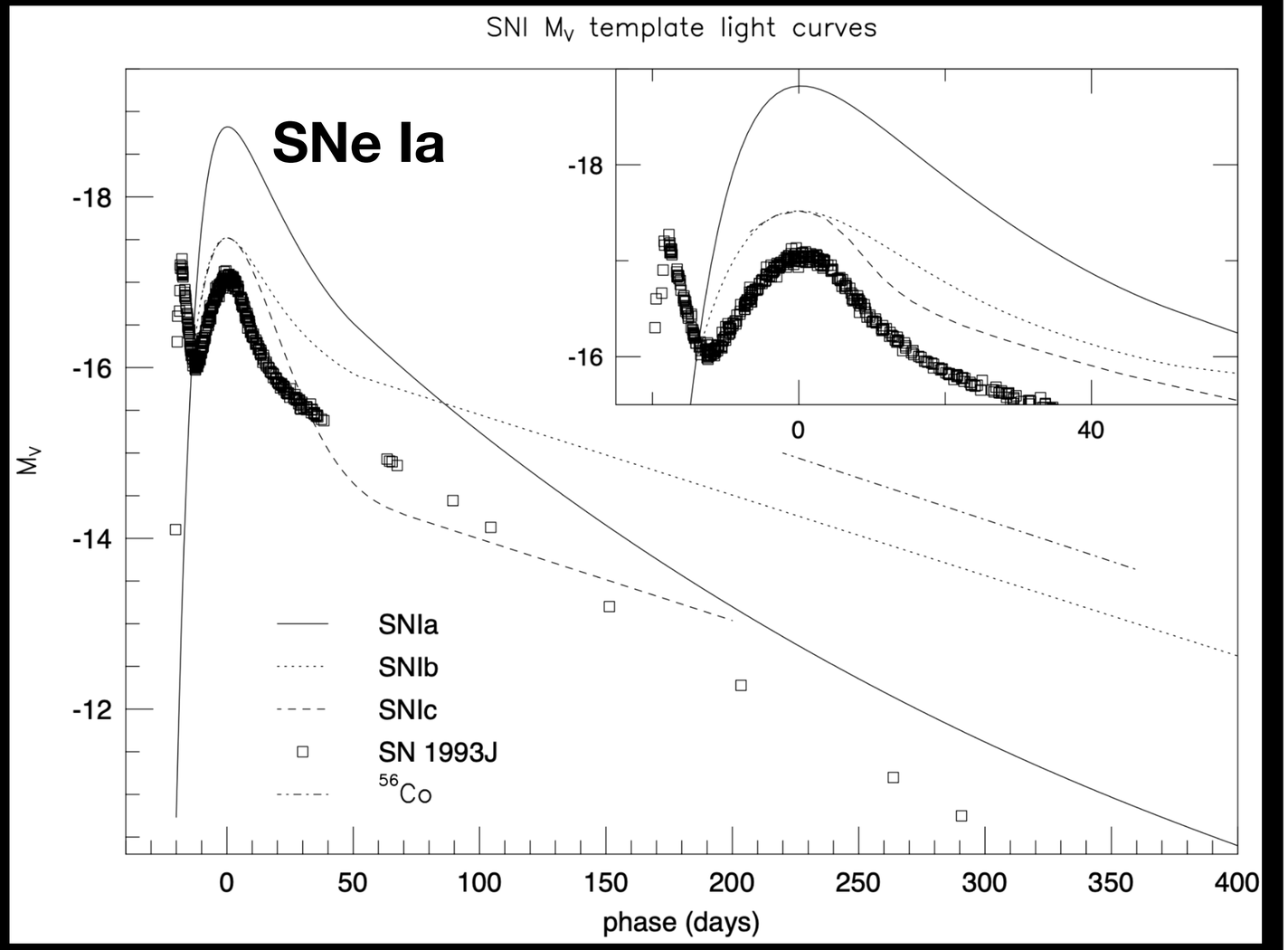


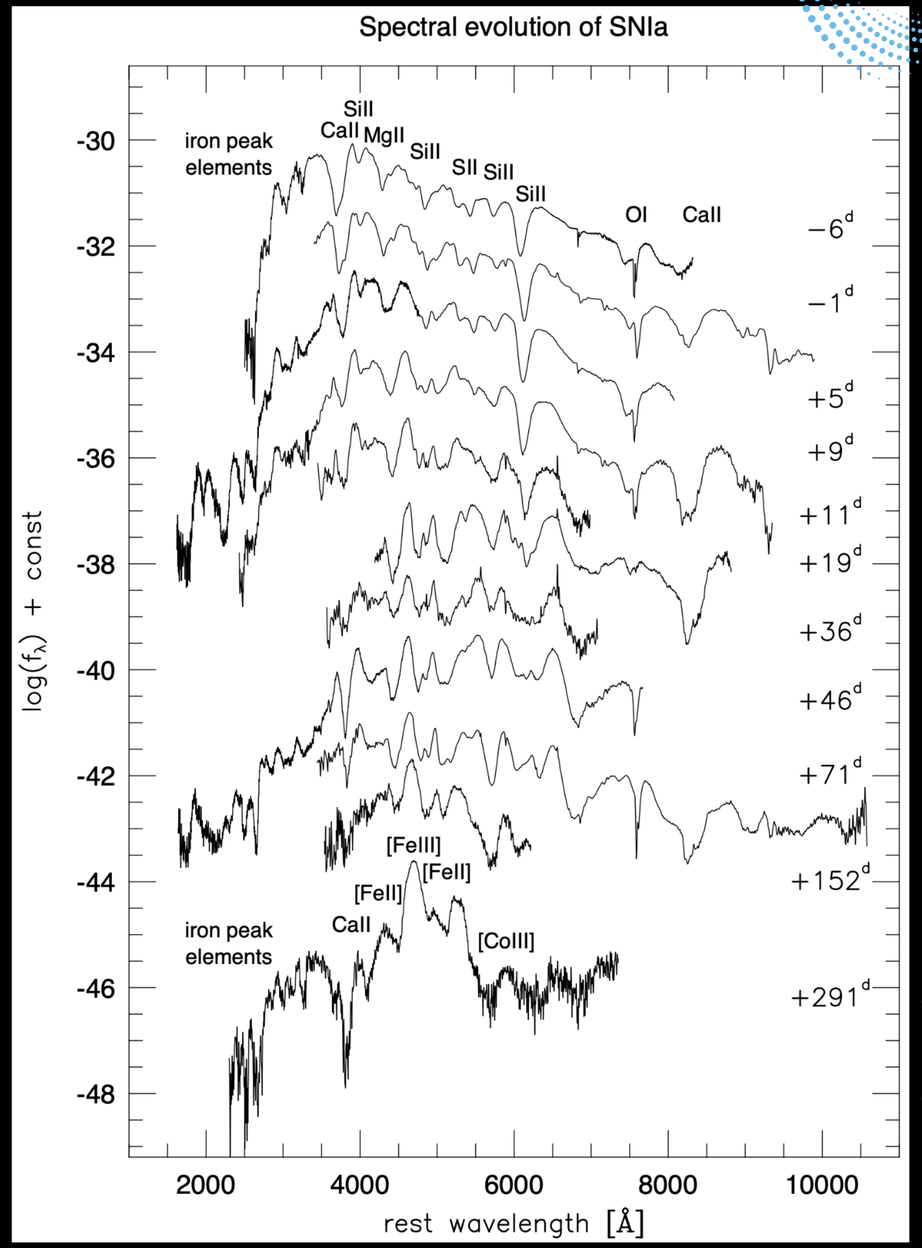
Kasliwal (2011)



Adapted from Turatto (2003)

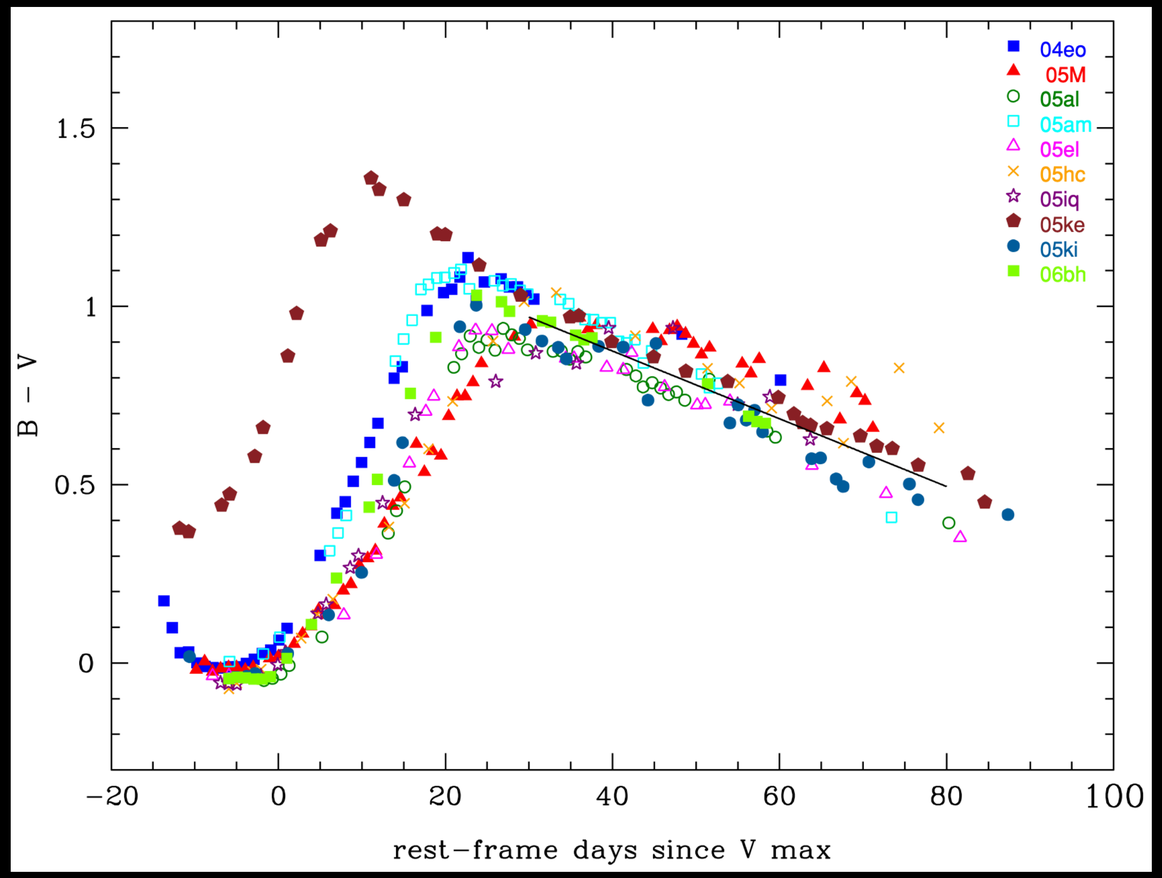
Thermonuclear SNe (Type Ia)





Wheeler, Benetti (2000)

The Lira relation



Folatelli+ (2010)

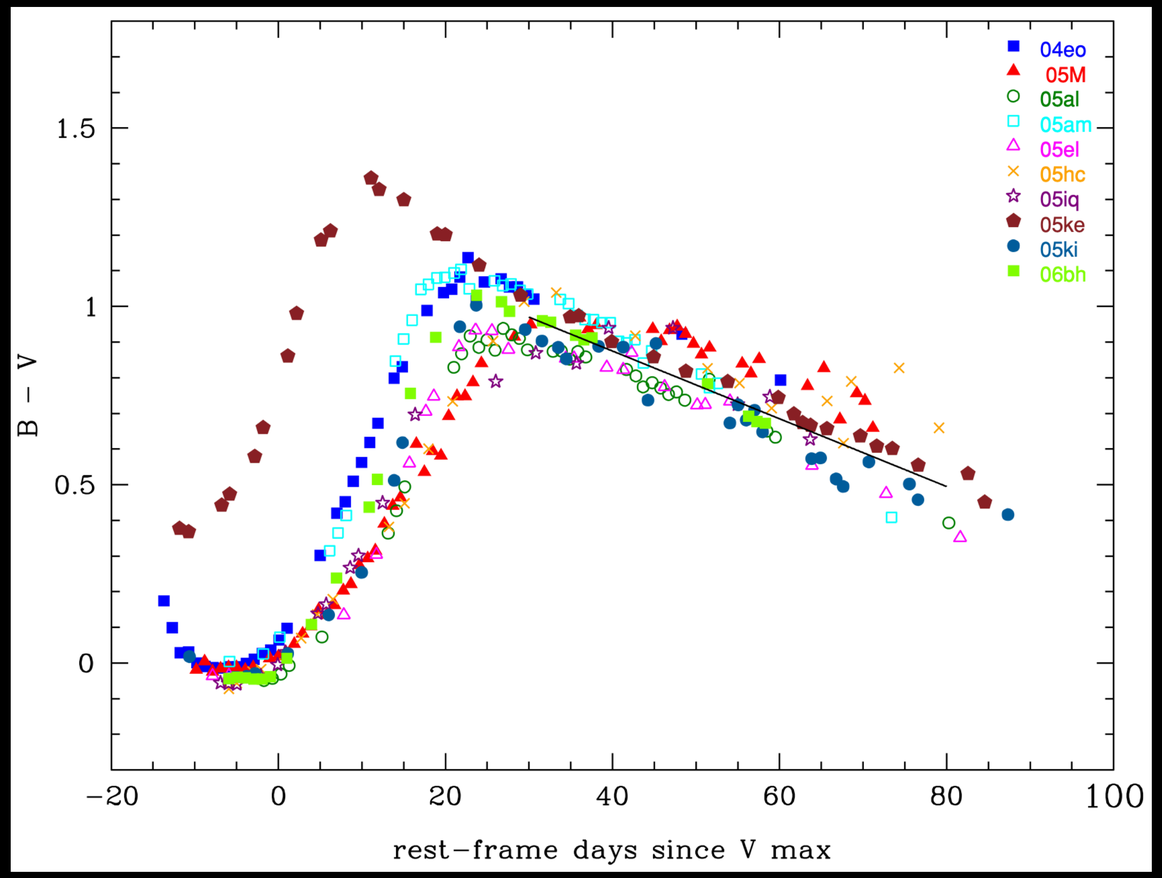
$$(B-V)_0 = A + Bt_*$$

$$A = 0.732 \pm 0.006$$

$$B = -0.0095 \pm 0.0005$$

$$t_* = t - t_{V_{max}} - 55$$

The Lira relation



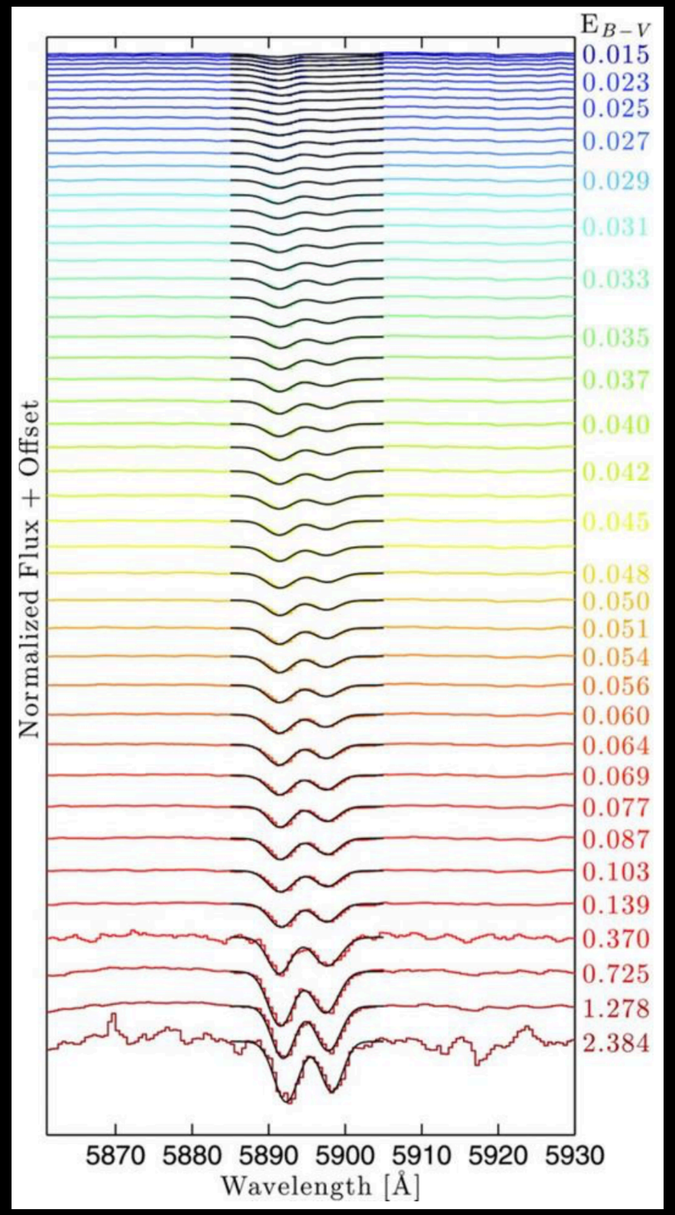
Folatelli+ (2010)

$$(B-V)_0 = A + Bt_*$$

“During the time period where the Lira relation holds (≈30–60 days after maximum light) the spectra […] form in the regions with iron-group elements, which have similar density structures and abundances.”

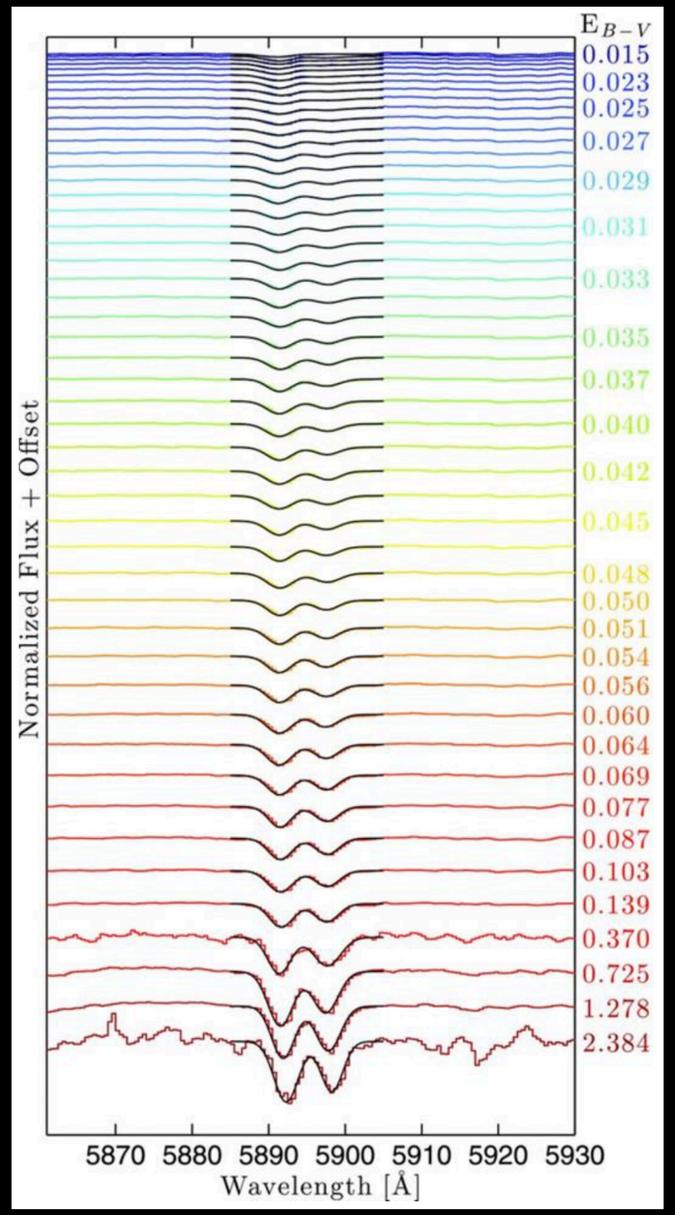
Hoeflich + (2017)

Na ID as extinction tracer

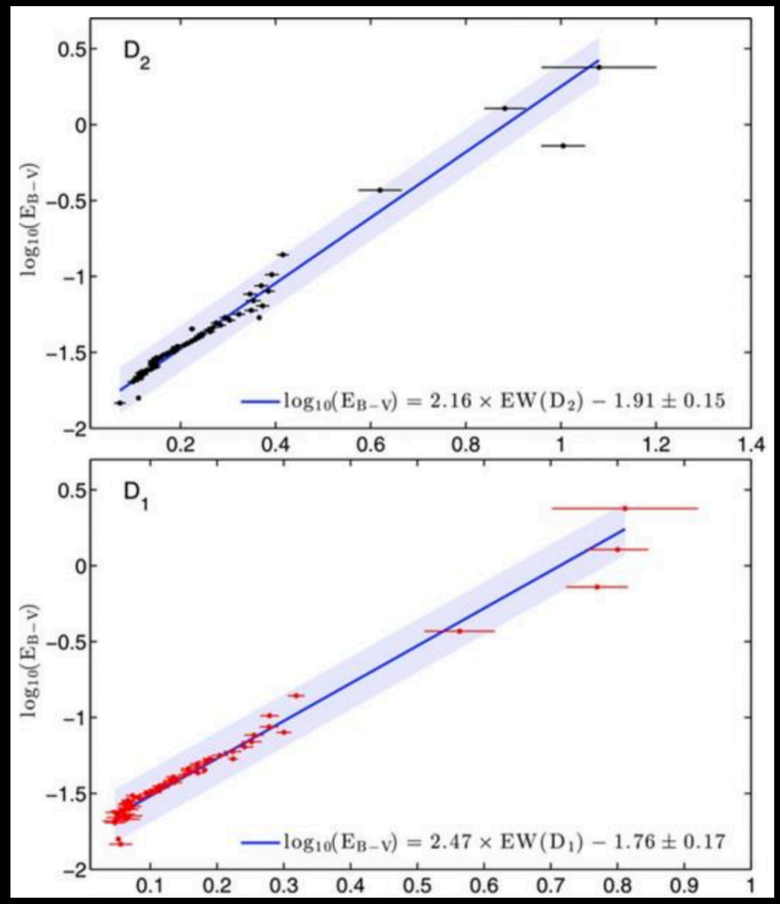


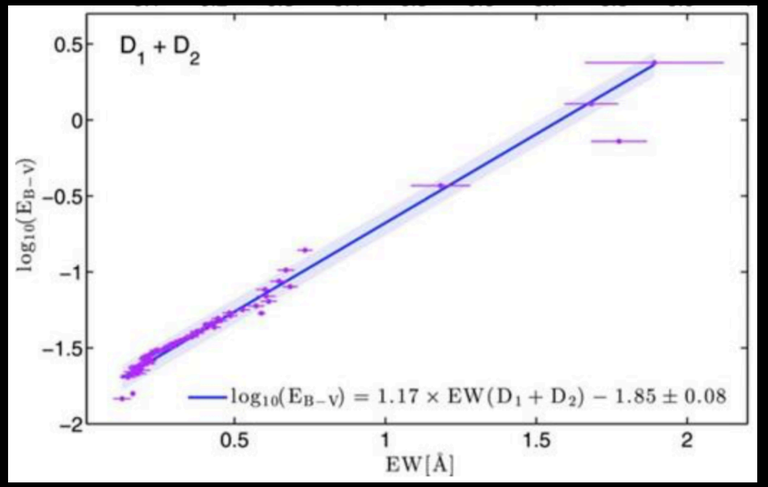
Poznanski+ (2012)

Na ID as extinction tracer



Poznanski+ (2012)





Na ID vs SNe Ia

Turatto+ (2003)

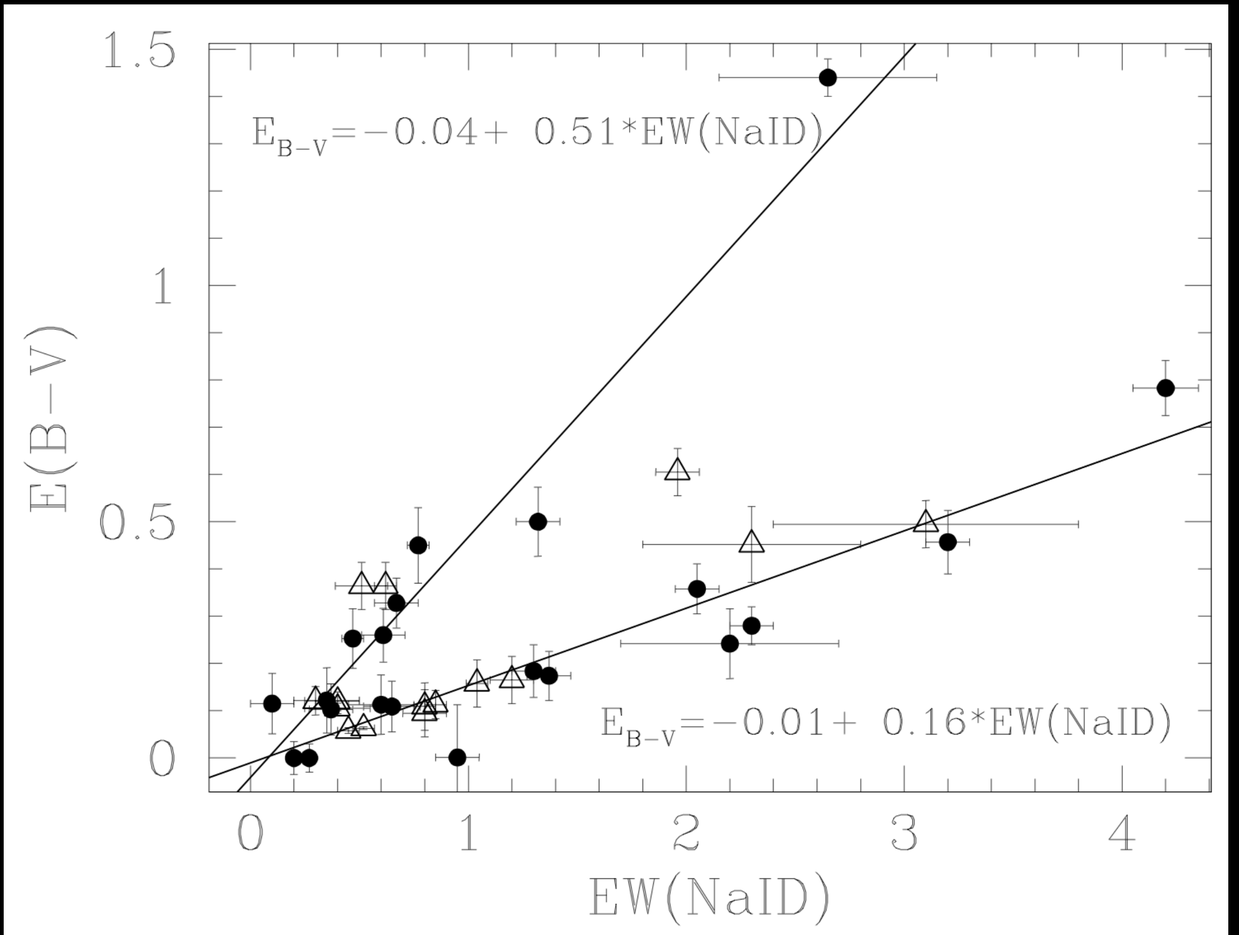
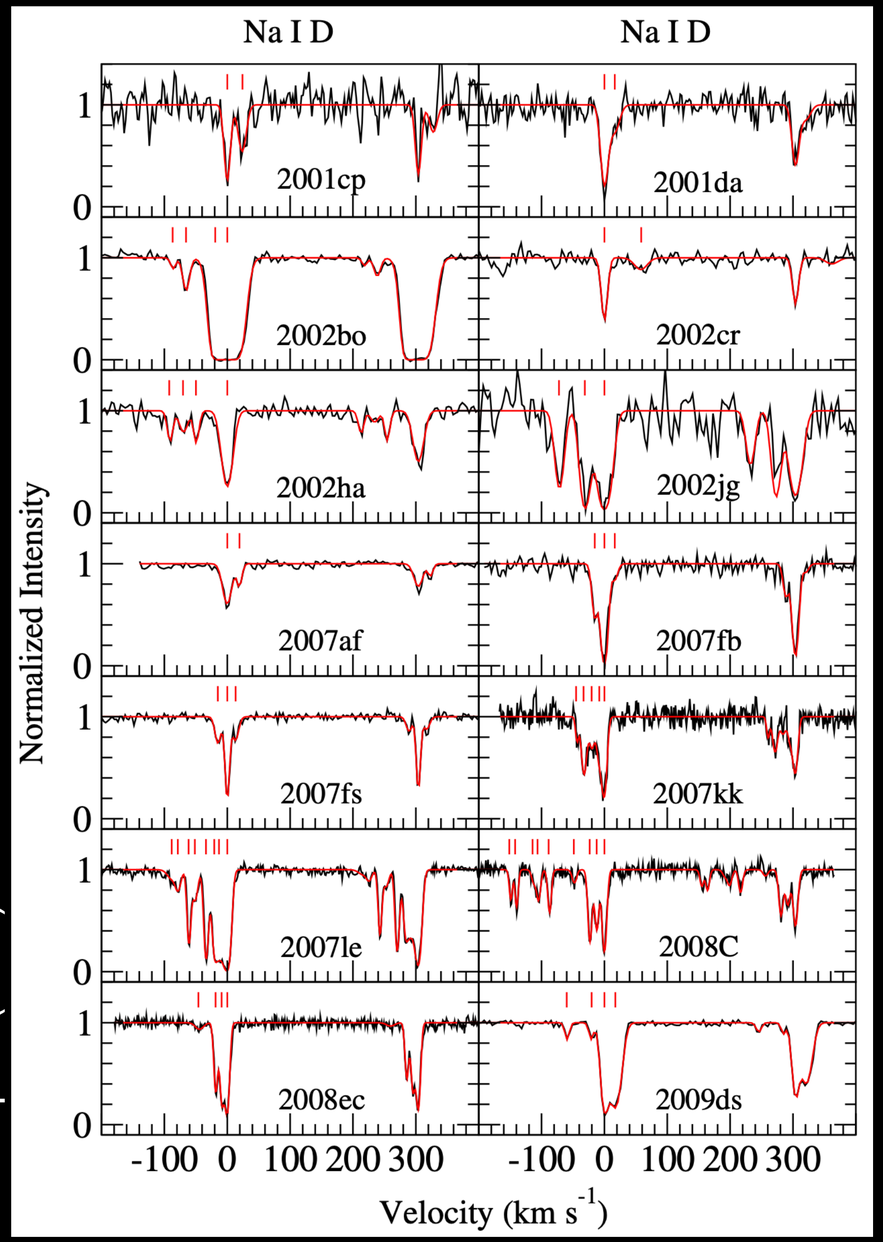


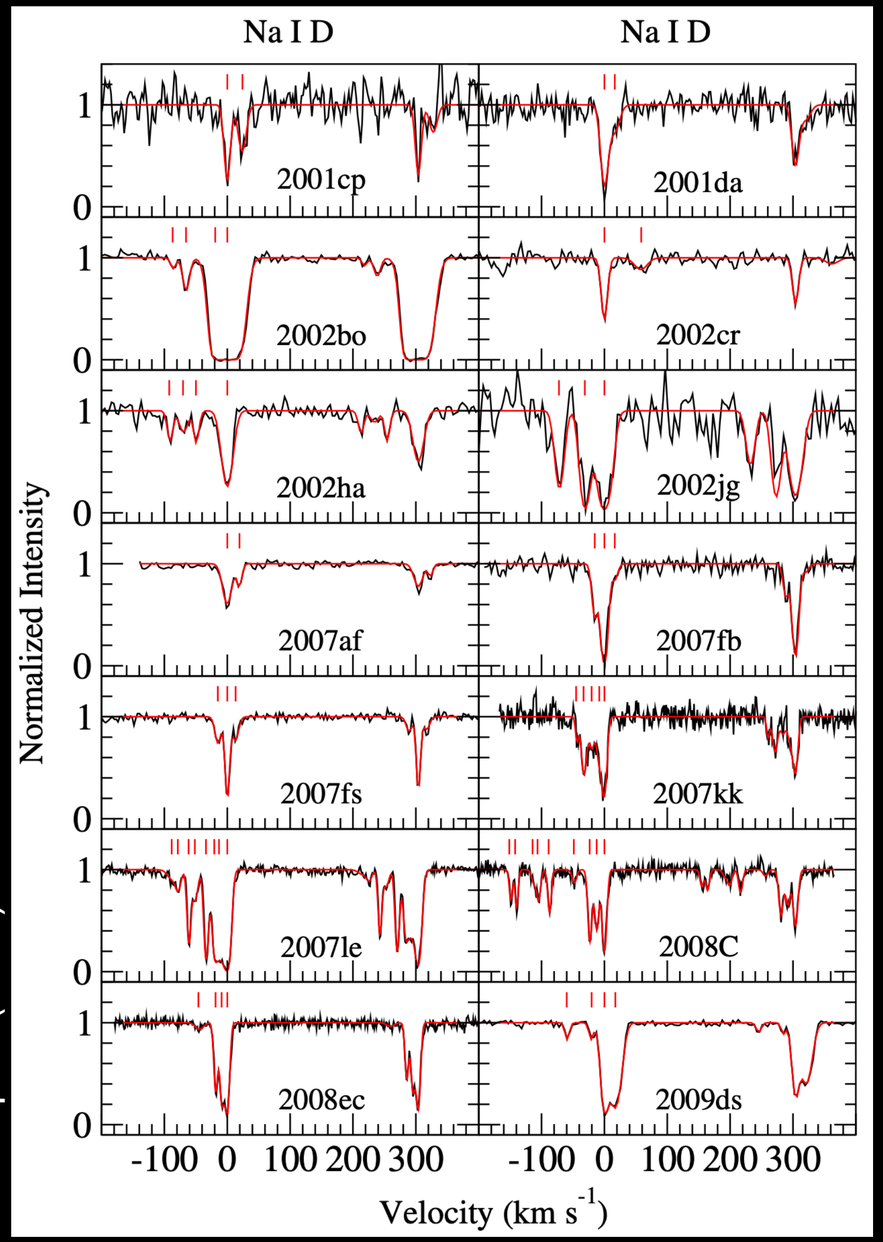
Fig. 3. Color excess of SNIa inside the host galaxy, E(B–V), determined from the tail of the color curves [6], versus EW(NaID) measured on low dispersion spectra (filled circles). The values of SN 1986G (the rightmost point) include also the Galactic component. Open triangles are estimates of Galactic color excess from for SNe of all types [15]

Na ID vs SNe Ia

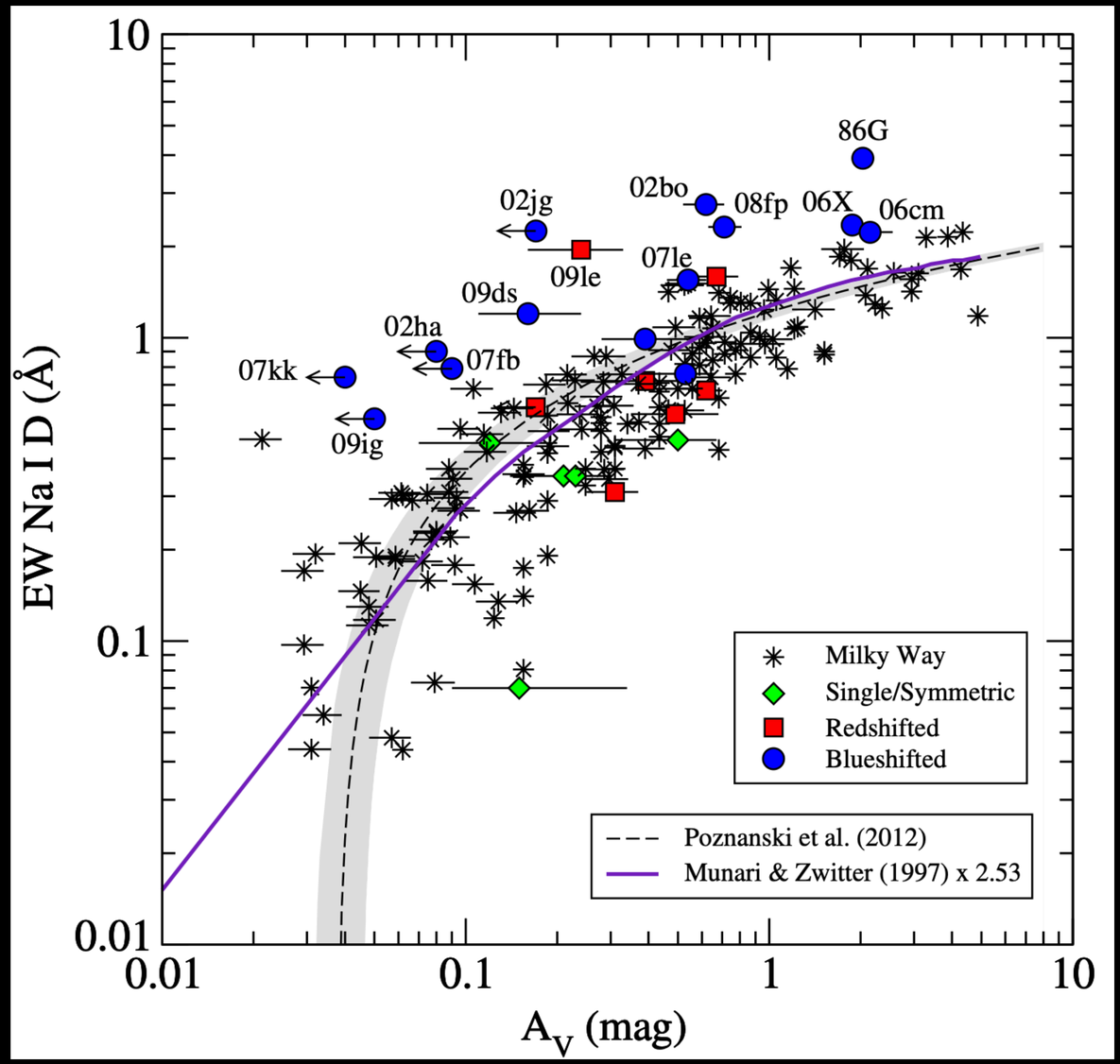


Phillips+ (2013)

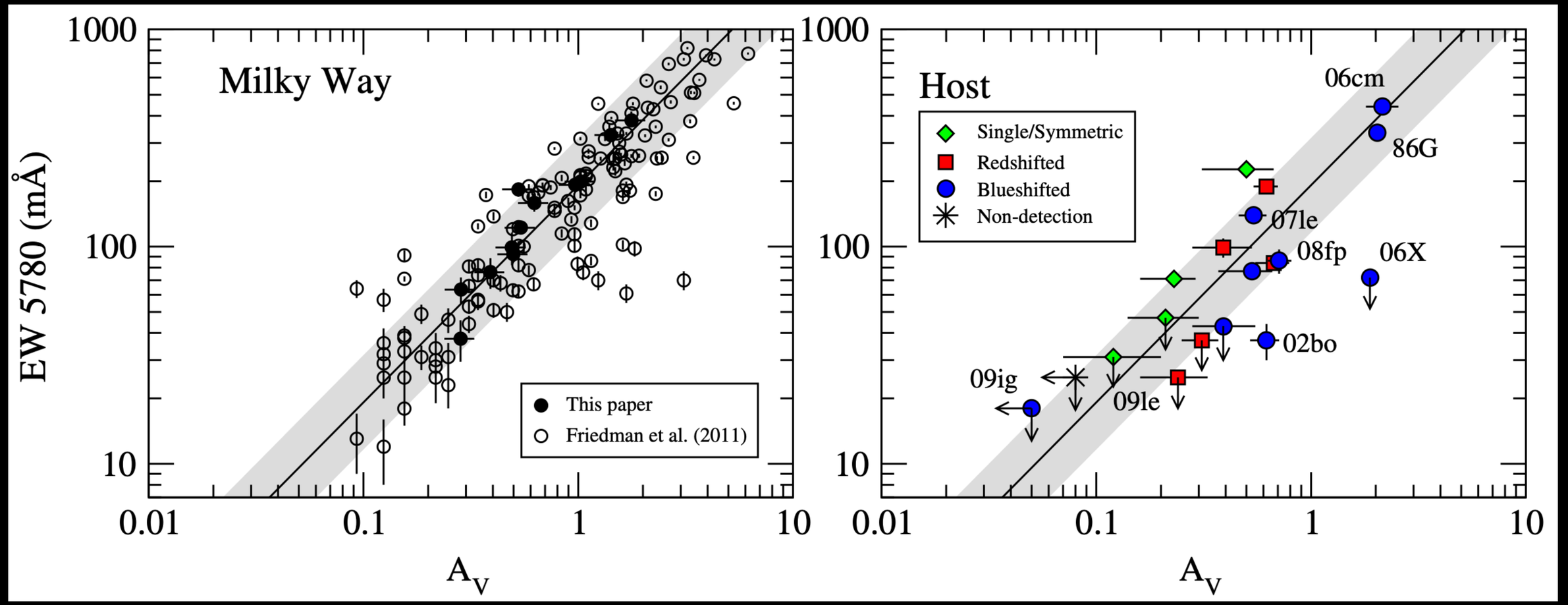
Na ID vs SNe Ia



Phillips+ (2013)



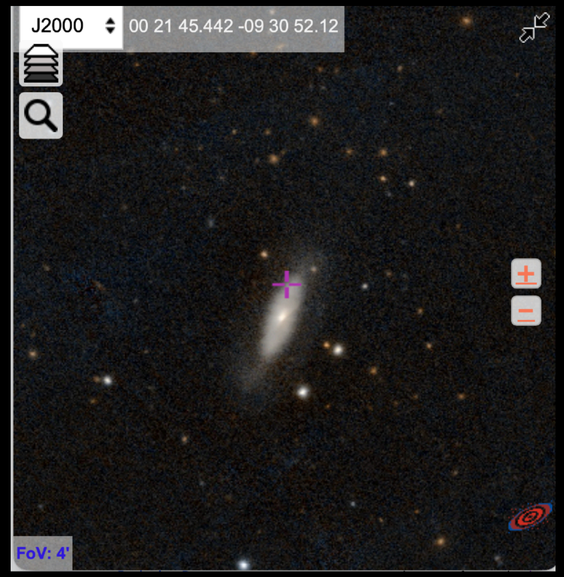
DIBS



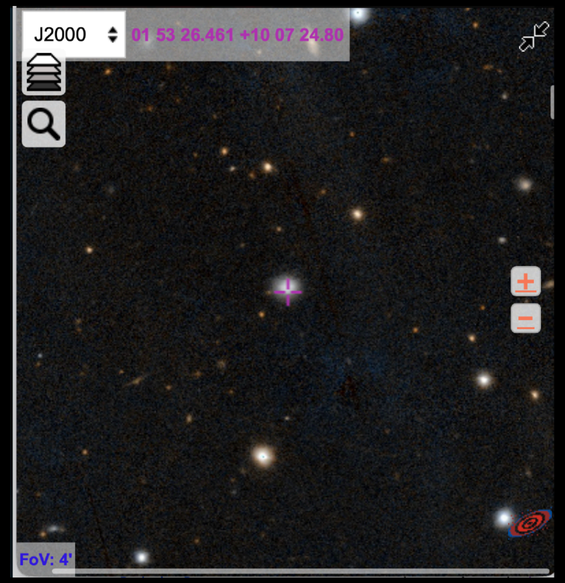
Phillips+ (2013)

Our Targets

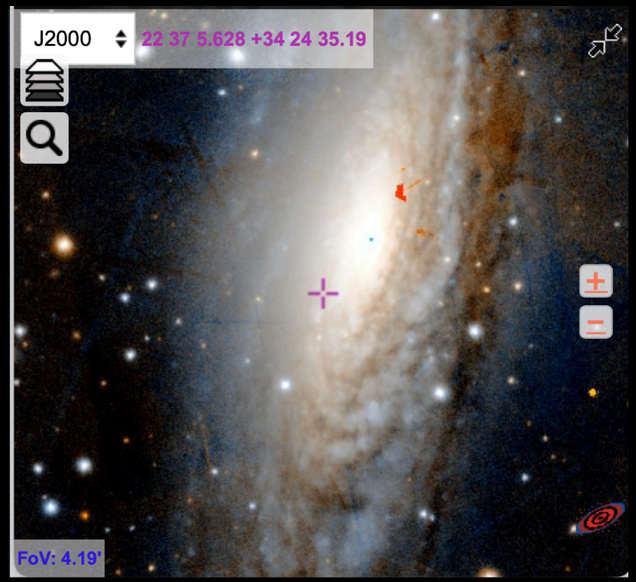
SN 2026zwp



SN 2026aaiv



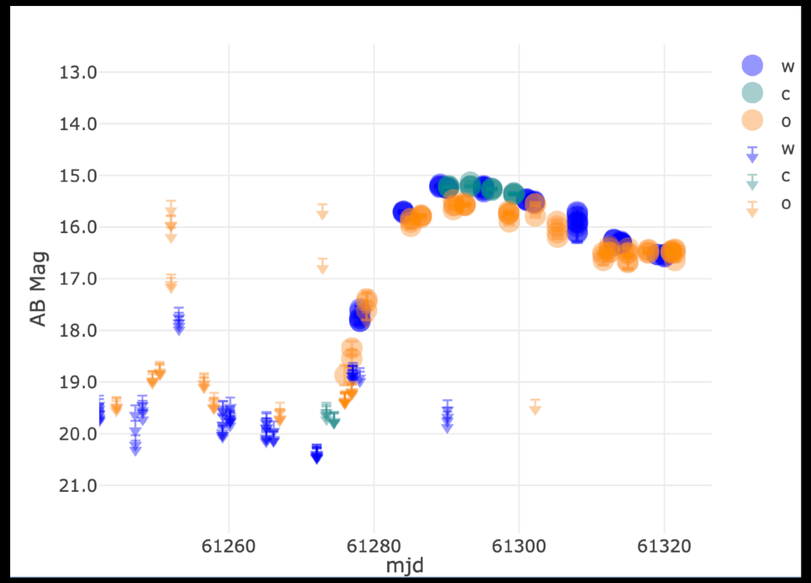
SN 2026aacj



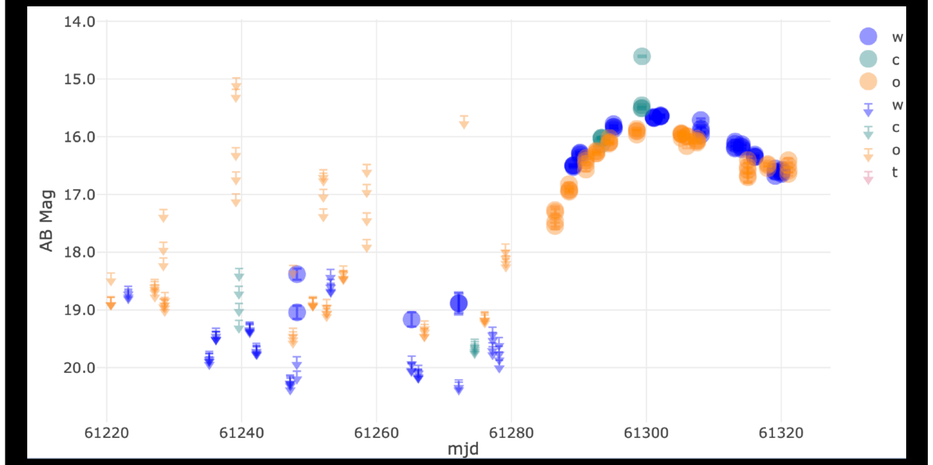
Our Targets

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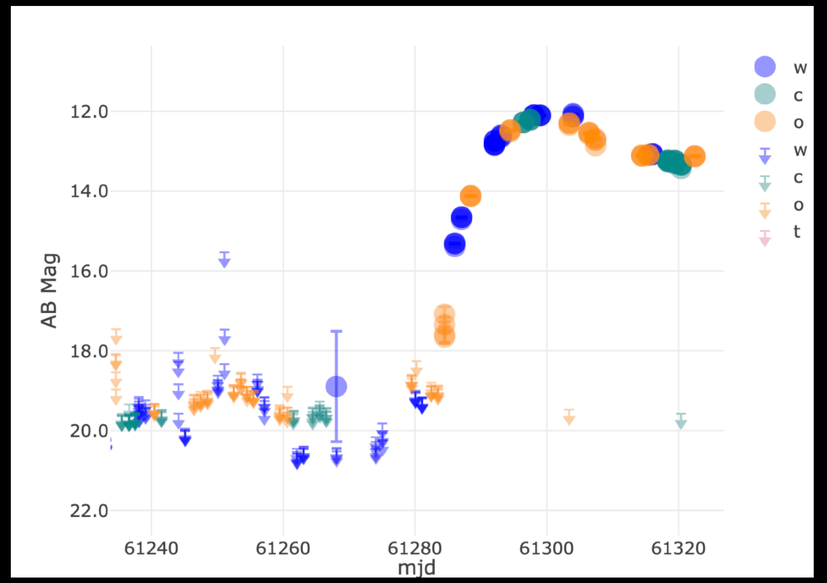
SN 2026zwp



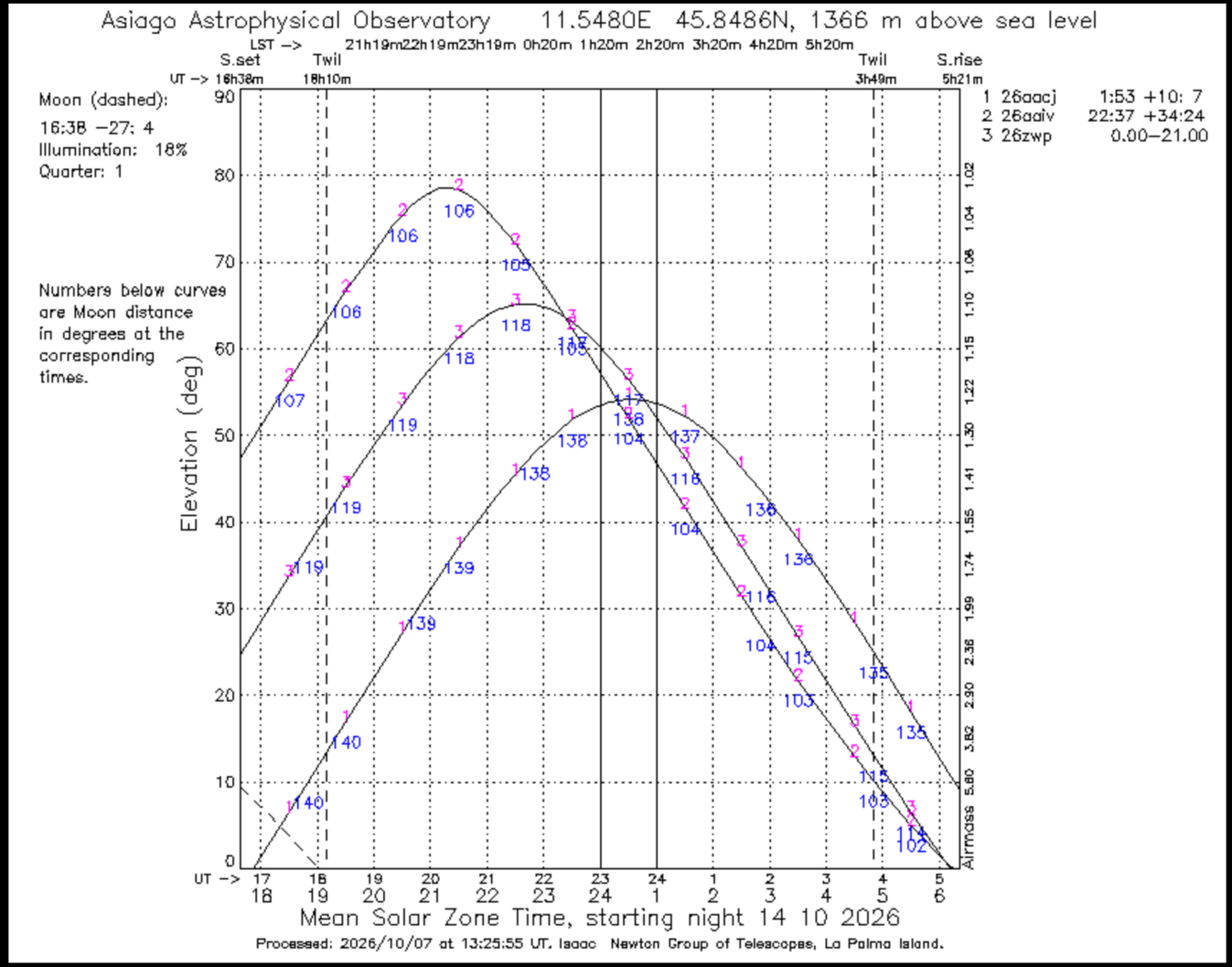
SN 2026aaiv



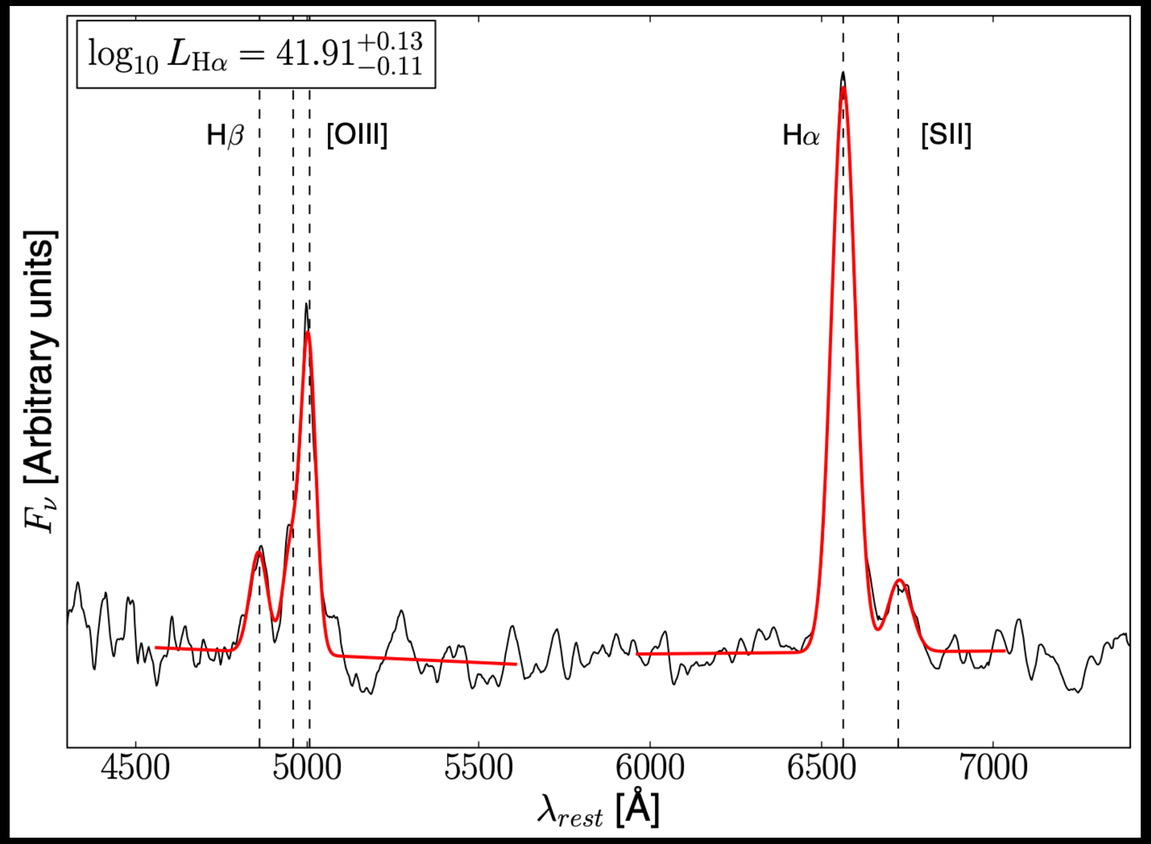
SN 2026aacj



Our Targets


Asiago Astrophysical Observatory 11.5480E 45.8486N, 1366 m above sea level
LST -> 21h19m22h19m23h19m 0h20m 1h20m 2h20m 3h20m 4h20m 5h20m
S.set Twil Twil S.rise
UT -> 16h38m 18h10m 3h49m 5h21m
Moon (dashed):
16:38 −27: 4
Illumination: 18%
Quarter: 1
Numbers below curves are Moon distance in degrees at the corresponding times.
1 26aacj 1:53 +10: 7
2 26aaiv 22:37 +34:24
3 26zwp 0.00−21.00
Elevation (deg)
Airmass
UT -> 17 18 19 20 21 22 23 24 1 2 3 4 5
18 19 20 21 22 23 24 1 2 3 4 5 6
Mean Solar Zone Time, starting night 14 10 2026
Processed: 2026/10/07 at 13:25:55 UT. Isaac Newton Group of Telescopes, La Palma Island.

Bonus: the Balmer Decrement



The intrinsic Balmer decrement remains roughly constant for typical gas conditions in star-forming galaxies (see Osterbrock 1989). In our analysis, we assume the value of $(\mathrm{H}\alpha/\mathrm{H}\beta)_{\mathrm{int}} = 2.86$, corresponding to a temperature $T = 10^4$ K and an electron density $n_e = 10^2$ cm^{-3} for Case B recombination (Osterbrock 1989). This choice is standard for star-forming galaxies in the literature. Thus, the nebular color excess is given by the following equation:

$$E(B-V) = 1.97 \log_{10} \left[\frac{(\mathrm{H}\alpha/\mathrm{H}\beta)_{\mathrm{obs}}}{2.86} \right]. \qquad (4)$$

Domínguez + (2013)